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## 10.2 Case Study: Population Estimation Using Landsat ETM+ Imagery

### 10.2.1 Study Area and Datasets

The city of Indianapolis, Indiana, has been chosen to implement this study. A detailed description and a map of the study area can be found in Sec. 3.2.1 of Chap. 3. A landsat 7 ETM+ image (row/path: 32/21) dated June 22, 2000, was used in this research. Atmospheric conditions were clear at the time of image acquisition, and the image was acquired through the U.S. Geological Survey's (USGS's) Earth Resource Observation Systems Data Center, which had corrected the radiometric and geometric distortions of the image to a quality level of 1G before delivery.

Population data at block group level were obtained from an ESRI Data and Maps CD, which was provided by the ESRI Company, based on a combination of topologically integrated geographic encoding and referencing (TIGER) files and 2000 U.S. Census population data. Because of different coordinate systems used for the Census data and the ETM+ image, the geographic coordinates of the Census data were converted to the Universal Transverse Mercator (UTM) to match with those of the ETM+ image.

### 10.2.2 Methods

#### Principal Component Analysis

Remotely sensed data, such as visible bands in Landsat TM/ETM+ images, are highly correlated between the adjacent spectral bands (Barnsley, 1999). Several techniques have been developed to transform highly correlated bands into an orthogonal subset. PCA is used most commonly. After performing PCA, the original correlated bands are transformed into independent principal components (PCs), of which the first PC contains the largest portion of data variance, and the second PC contains the second largest data variance, and so on. The higher-numbered PCs often appear noisy because they contain very little variance of information (Richards, 1994). In this study, six ETM+ multispectral bands (1 to 5 and 7) were used to perform PCA. The first three PCs were used in population-estimation analysis because they accounted for 99 percent of total variance.

#### Vegetation Indices

Many vegetation indices have been developed based on the fact that plants reflect less in visible red light but more in near-infrared radiation than nonvegetated surfaces (Bannari et al., 1995; Jensen, 2000). Thus vegetation indices can enhance or extract some specific features

| Vegetation Index                         | Abbr. | Formula                                                                          | References              |
|------------------------------------------|-------|----------------------------------------------------------------------------------|-------------------------|
| Normalized difference vegetation index   | NDVI  | $\frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}$                        | Rouse et al., 1974      |
| Soil adjusted vegetation index           | SAVI  | $\frac{(1+L)(\text{NIR} - \text{RED})}{\text{NIR} + \text{RED} + L}$ , $L = 0.5$ | Huete, 1988             |
| Renormalized difference vegetation index | RDVI  | $\frac{\text{NIR} - \text{RED}}{\sqrt{\text{NIR} + \text{RED}}}$                 | Roujean and Breon, 1995 |
| Transformed NDVI                         | TNDVI | $\sqrt{\text{NDVI} + 0.5}$                                                       | Deering et al., 1975    |
| Simple vegetation index                  | SVI   | $\text{NIR} - \text{RED}$                                                        |                         |
| Simple ratio                             | RVI   | $\text{NIR} / \text{RED}$                                                        | Birth and McVey, 1968   |

Note: NIR = near-infrared wavelength, ETM+ band 4; RED = red wavelength, ETM+ band 3.

**TABLE 10.2** Definition of Vegetation Indices Used

that single spectral bands cannot. In this research, six vegetation indices, namely, the normalized difference vegetation index (NDVI), the soil adjusted vegetation index (SAVI), the renormalized difference vegetation index (RDVI), the transformed NDVI (TNDVI), the simple vegetation index (SVI), and the simple ratio (RVI), were examined to use for population estimation (Table 10.2).

**Fraction Images**

Spectral mixture analysis (SMA) is regarded as a physically based image processing tool that supports repeatable and accurate extraction of quantitative subpixel information (Mustard and Sunshine, 1999; Roberts et al., 1998; Smith et al., 1990). It assumes that the spectrum measured by a sensor is a linear combination of the spectra of all components within the pixel (Adams et al., 1995; Roberts et al., 1998). Because of its effectiveness in handling spectral mixture problems, SMA has been used widely in estimation of vegetation cover (Asner and Lobell, 2000; McGwire et al., 2000; Small, 2001; Smith et al., 1990), in vegetation or land-cover classification and change detection (Adams et al., 1995; Aguiar et al., 1999; Cochrane and Souza, 1998; Lu et al., 2003; Roberts et al., 1998), and in urban studies (Phinn et al.,

2002; Rashed et al., 2001; Small, 2001; Wu and Murray, 2003). In this study, SMA was used to develop green-vegetation and impervious-surface fraction images. Endmembers were identified initially from the ETM+ image based on high-resolution aerial photographs. The shade endmember was identified from the areas of clear and deep water, whereas green vegetation was selected from the areas of dense grass and cover crops. Different types of impervious surfaces were selected, from building roofs to highway intersections. An unconstrained least-squares solution was used to decompose the six ETM+ bands (1 through 5 and 7) into three fraction images (e.g., vegetation, impervious surface, and shade). The fractions represent the areal proportions of the endmembers within a pixel. The shade fraction was not used owing to its irrelevance to the population distribution. A detailed description of this procedure can be found in Lu and Weng (2004).

### Texture Images

*Texture* often refers to the pattern of intensity of variations in an image. Many texture measures have been developed (Haralick, 1979; Haralick et al., 1973; He and Wang, 1990) and used for land-cover classification (Gong and Howarth, 1992; Marceau et al., 1990; Narasimha Rao et al., 2002; Shaban and Dikshit, 2001). A common texture measure, variance, has been shown to be useful in improving land-cover classification (Shaban and Dikshit, 2001). In this study, variance was developed and used to examine its relationship with population. Landsat ETM+ bands 3 and 7, which correlate strongly with urban features, were used for deriving texture images with window sizes of  $3 \times 3$ ,  $5 \times 5$ , and  $7 \times 7$ .

### Temperature

A surface-temperature image was extracted from the ETM+ thermal infrared band (band 6). The procedure to develop the surface temperature involves three steps: (1) converting the digital number of ETM+ band 6 into spectral radiance, (2) converting the spectral radiance to at-satellite brightness temperature, which is also called *black-body temperature*, and (3) converting the blackbody temperature to land surface temperature. A detailed description of how the temperature image was developed can be found in Weng and colleagues (2004).

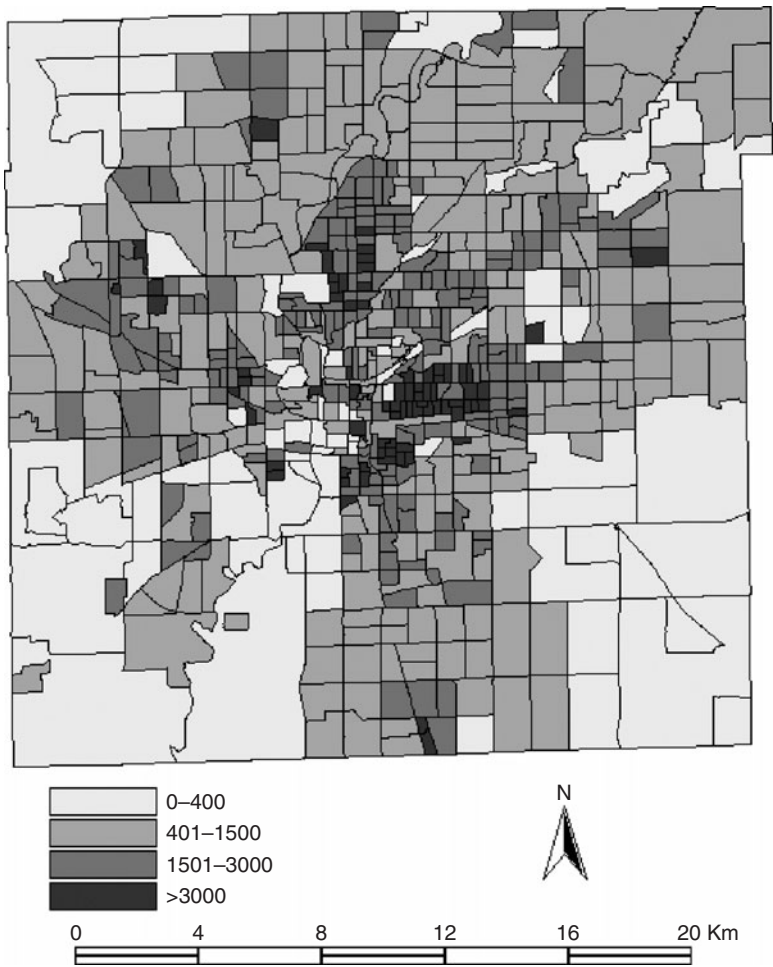
### Model Development

Since Census data and ETM+ data have different formats and spatial resolutions, they need to be integrated. With the help of ERDAS IMAGINE, remotely sensed data were aggregated to block-group level. The mean values of selected remote sensing variables at the block-group level were computed. The variables include radiances of ETM+ bands, principal components, vegetation indices, green-vegetation

and impervious surface fractions, temperatures, and texture indicators. All these data then were exported into SPSS software for correlation and regression analysis.

Twenty-five percent of the total block groups (658) in the city were randomly selected. A 2.5 standard deviation was used to identify the outliers. A total of 162 samples was used for developing models with a non-stratified sampling scheme. The population density in Indianapolis was calculated to range from 0 to 7253 persons/km<sup>2</sup>, whereas most block groups had a population density that ranged from 400 to 3000 persons/km<sup>2</sup> (Fig. 10.1).

Previous research has indicated that extremely high or low population density is difficult to estimate using remotely sensed data



**FIGURE 10.1** Population-density distribution by block groups in Indianapolis based on the 2000 Census.

| Category       | Samples            | Min. | Max. | Mean    | SD     |
|----------------|--------------------|------|------|---------|--------|
| Non-stratified | 162* (175†) (658‡) | 8    | 4479 | 1470.71 | 948.62 |
| Low            | 77* (82†)          | 1    | 393  | 208.94  | 123.11 |
| Medium         | 114* (125†) (499‡) | 402  | 2824 | 1417.31 | 676.97 |
| High           | 70* (77†)          | 3015 | 5189 | 3707.66 | 579.08 |

\*Samples with outliers removed that finally were used for data analysis.

†Samples selected based on a random sampling technique.

‡Total number of block groups corresponding to population.

**TABLE 10.3** Statistical Descriptions of Samples of Population Densities (persons/km<sup>2</sup>)

(Harvey, 2002a, 2000b; Lo, 1995); hence the population densities of the city were divided into three categories: low (fewer than 400 persons/km<sup>2</sup>), medium (401 to 3000 persons/km<sup>2</sup>), and high (more than 3000 persons/km<sup>2</sup>) based on the data distribution. All block groups in the low- and high-density categories were used for sampling owing to their limited number. For the medium-density category, samples were chosen using a random sampling technique. Table 10.3 summarizes the statistical characteristics of selected samples for different categories.

Pearson's correlation coefficients were computed between population densities and the remote sensing variables. Stepwise regression analysis was further applied to identify suitable variables for developing population-estimation models. The coefficient of determination ( $R^2$ ) was used as an indicator to determine the robustness of a regression model. To improve model performance, various combinations of the remote sensing variables were explored, as well as the transformation of population densities (PD) into natural-logarithm (LPD) and square-root (SPD) forms.

### Accuracy Assessment

Whenever a model is applied for prediction, there are always discrepancies between true and estimated values, and these are called *residuals*. It is necessary to validate whether the model fits training-set data, which is called *internal validation*, or to test its fitness with other data sets that are not used as training sets, which is called *external validation* (Harvey, 2002a). Relative and absolute error can be computed. For an individual case, the relative error can be expressed as

$$RE = (P_g - P_e)/P_g \times 100 \quad (10.2)$$

where  $P_g$  and  $P_e$  are the reference and estimated values, respectively. The residual ( $P_g - P_e$ ) for individual cases may be negative or positive,

so absolute values of the residuals are used to assess the overall performance of the model; that is,

$$\text{Overall relative error (RE)} = \frac{\sum_{k=1}^n |\text{RE}_k|}{n} \quad (10.3)$$

$$\text{Overall absolute error (AE)} = \frac{\sum_{k=1}^n |P_g - P_e|}{n} \quad (10.4)$$

where  $n$  is the number of block groups used for accuracy assessment. The smaller the RE and AE, the better the models will be. A total of 483 unsampled block groups was used to assess the performance of models in the non-stratified sampling scheme. For the stratified sampling scheme, a total of 521 samples was used for accuracy assessment. A residual map was created based on the best estimation model for geographic analysis of predicted errors.

### 10.2.3 Result of Population Estimation Based on a Non-Stratified Sampling Method

Six groups of remote sensing variables were used to explore their relationship with population parameters, and their correlation coefficients are presented in Table 10.4.

Table 10.4 indicates that among the ETM+ spectral bands, band 4 was the most strongly correlated with population density; the transforms of population density into natural-logarithm or square-root forms did not improve the correlation coefficients of single ETM+ bands except for band 5. The principle components, especially PC2, improved the correlation with population parameters when compared with single ETM+ bands. All selected vegetation indices had a significant correlation with population density. The green-vegetation fraction had a better correlation with population density than the impervious-surface fraction. Selected textures, especially band 7 associated with a window size of  $7 \times 7$ , were strongly correlated with population density. Among all selected remote sensing variables, temperature was the most correlated variable with population density. Moreover, it was found that vegetation-related variables such as band 4, PC2, vegetation indices, and the green-vegetation fraction all had a negative correlation with population parameters. This is so because for a given area, more vegetation is often related to less built-up area and thus less population.

The strong correlations between population parameters and several remote sensing variables imply that a combination of temperature, textures, and spectral responses could be used to improve the population-estimation models. A series of estimation models was developed by performing stepwise regression analysis based on different

| Variables |          | PD      | SPD     | LPD     |
|-----------|----------|---------|---------|---------|
| Bands     | B1       | 0.226*  | 0.160†  | 0.019   |
|           | B2       | 0.163†  | 0.096   | -0.039  |
|           | B3       | 0.164†  | 0.096   | -0.039  |
|           | B4       | -0.255* | -0.209* | -0.108  |
|           | B5       | -0.155† | -0.196† | -0.251* |
|           | B7       | 0.068   | 0.003   | -0.115  |
| PCs       | PC1      | 0.123   | 0.056   | -0.073  |
|           | PC2      | -0.319* | -0.283* | -0.190† |
|           | PC3      | -0.248* | -0.239* | -0.178† |
| VIs       | NDVI     | -0.244* | -0.182† | -0.052  |
|           | RDVI     | -0.242* | -0.178† | -0.040  |
|           | SAVI     | -0.245* | -0.182† | -0.053  |
|           | SVI      | -0.221* | -0.156† | -0.023  |
|           | RVI      | -0.385* | -0.337* | -0.206* |
|           | TNDVI    | -0.164† | -0.098  | 0.026   |
| Frac.     | GV       | -0.231* | -0.171† | -0.045  |
|           | IMP      | 0.109   | 0.043   | -0.082  |
| Text.     | B3_3 × 3 | -0.196† | -0.223* | -0.267* |
|           | B7_3 × 3 | -0.295* | -0.326* | -0.347* |
|           | B3_5 × 5 | -0.280* | -0.317* | -0.360* |
|           | B7_5 × 5 | -0.368* | -0.406* | -0.427* |
|           | B3_7 × 7 | -0.322* | -0.364* | -0.407* |
|           | B7_7 × 7 | -0.402* | -0.444* | -0.463* |
| Temp.     | TEMP     | 0.519*  | 0.513*  | 0.411*  |

Note: Bn = band n; PD = population density; SPD = square root of population density; LPD = natural logarithm of population density; PCs = principal components; VIs = vegetation indices; Frac. = fraction images; GV = green-vegetation fraction; IMP = impervious surface fraction; Text. = texture; Temp. = temperature.

\*Correlation at 99 percent confidence level (two-tailed).

†Correlation at 95 percent confidence level (two-tailed).

**TABLE 10.4** Relationships between Population Parameters and Remote Sensing Variables Based on Non-Stratified Samples

combinations of remote sensing variables. The predictors and  $R^2$  of the regression models developed are presented in Table 10.5.

Table 10.5 indicates that any single group of remote sensing variables did not produce a satisfactory  $R^2$  except for vegetation indices. Incorporation of vegetation-related variables or use of all variables provided better modeling results. The square-root form of population density improved the regression models, whereas the natural-logarithm form degraded the regression performance, with an exception in the textures. Table 10.6 summarizes the best-performing regression models and associated estimation errors.

| Potential Variables | PD                                                |       | SPD                                               |       | LPD                              |       |
|---------------------|---------------------------------------------------|-------|---------------------------------------------------|-------|----------------------------------|-------|
|                     | Selected Var.                                     | $R^2$ | Selected Var.                                     | $R^2$ | Selected Var.                    | $R^2$ |
| Bands               | B4                                                | 0.065 | B5, B1, B2                                        | 0.212 | B1, B2, B5                       | 0.160 |
| PCs                 | PC2, PC3                                          | 0.159 | PC2, PC3                                          | 0.134 | PC2, PC3, PC1                    | 0.107 |
| Vis                 | RVI, TNDVI, SAVI, RDVI                            | 0.622 | TNDVI, SAVI, RDVI, RVI                            | 0.645 | TNDVI, SAVI, RDVI                | 0.548 |
| Frac.               | GV, IMP                                           | 0.079 | GV, IMP                                           | 0.065 |                                  |       |
| Text.               | B7_7 $\times$ 7, B7_3 $\times$ 3, B3_3 $\times$ 3 | 0.369 | B7_7 $\times$ 7, B3_3 $\times$ 3, B3_5 $\times$ 5 | 0.465 | B7_3 $\times$ 3, B7_5 $\times$ 5 | 0.448 |
| Temp.               | TEMP                                              | 0.269 | TEMP                                              | 0.263 | TEMP                             | 0.169 |
| VRV                 | RVI, TNDVI, SAVI, PC2, B4                         | 0.768 | RVI, TNDVI, SAVI, PC2, B4                         | 0.797 | RVI, TNDVI, SAVI, PC2, B4        | 0.678 |
| B-temp.             | Temp., B5                                         | 0.351 | Temp., B7                                         | 0.376 | Temp., B7                        | 0.338 |
| Mixture             | B7_7 $\times$ 7, RVI, B2, TNDVI, SAVI, B5         | 0.785 | TEMP, RVI, TNDVI, SAVI, B5, RDVI, SVI             | 0.828 | TNDVI, SAVI, B5, TEMP, RVI       | 0.698 |

*Note:* VRV = vegetation-related variables, including band 4, PC2, VIs, and GV;  
 B-temp. = combination bands and temperature; Mixture = combination of all variables.

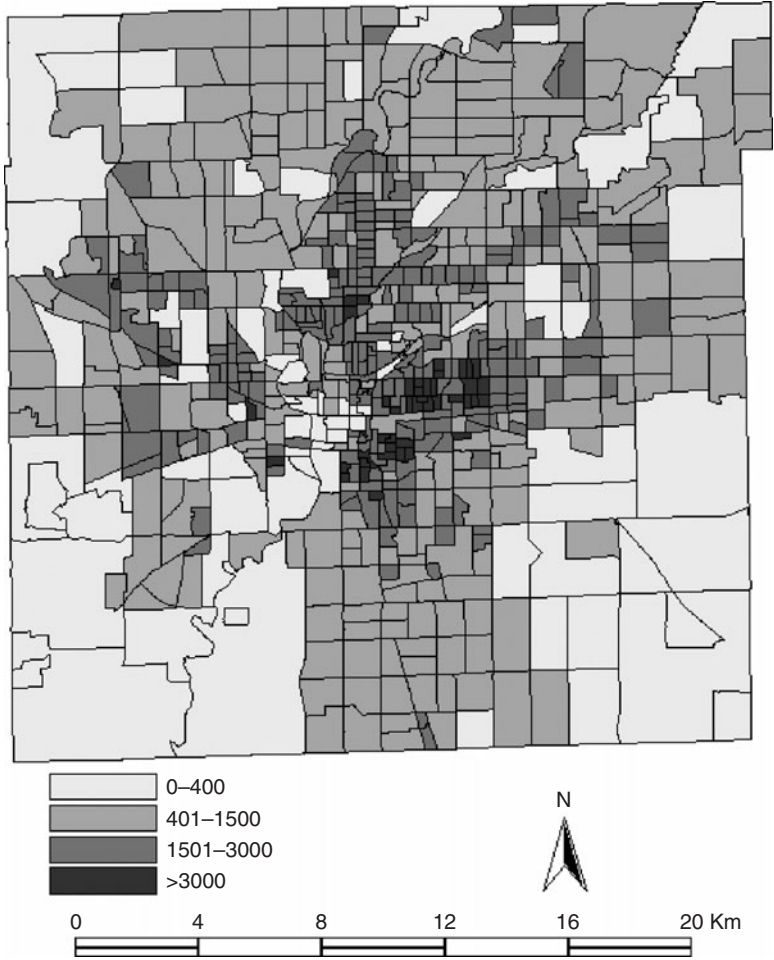
**TABLE 10.5** Comparison of Regression Results for Population-Density Estimation Based on Non-Stratified Samples



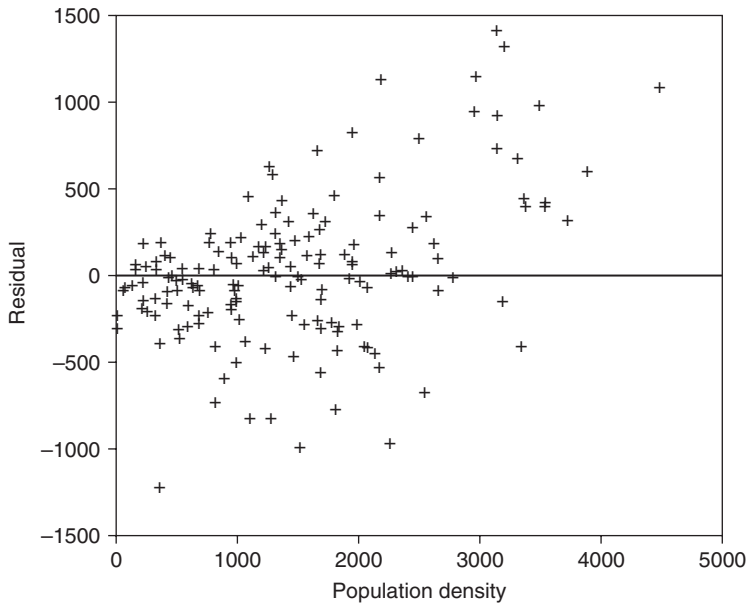
| Model | Variable | Regression Equation                                                                                                                                                   | $R^2$ | RE    | AE  |
|-------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-----|
| PD    | 1        | Mixture<br>$-83613.428 - 58.830 \times B7\_7 \times 7 + 5914.817 \times RVI + 117300.115 \times TNDVI - 65068.691 \times SAVI - 65.723 \times B5 + 64.369 \times B2$  | 0.785 | 204.3 | 505 |
|       | 2        | VRV<br>$-95394.477 + 6378.881 \times RVI + 132709.023 \times TNDVI - 73728.142 \times SAVI - 137.526 \times PC2 + 129.704 \times B4$                                  | 0.768 | 204.4 | 523 |
| SPD   | 3        | Mixture<br>$-1293.678 + 1.318 \times TEMP + 57.79 \times RVI + 1347.089 \times TNDVI - 789.683 \times SAVI - 1.124 \times B5 - 11.674 \times RDVI + 1.325 \times SVI$ | 0.828 | 123.1 | 439 |
|       | 4        | VRV<br>$-1226.463 + 72.752 \times RVI + 1754.789 \times TNDVI - 1.915 \times PC2 - 945.565 \times SAVI + 1.742 \times B4$                                             | 0.797 | 142.1 | 452 |

**TABLE 10.6** Summary of Selected Estimation Models for Population-Density Estimation Based on Non-Stratified Samples

Overall, larger  $R^2$  values resulted in fewer estimation errors. The regression models using a combination of spectral, texture, and temperature data provided the best estimation results. The  $R^2$  value for the best model (model 3) reached 0.83, but the estimation errors were still high. Figure 10.2 shows population-density distribution estimated using this model. The overall relative errors were larger than 123 percent, and the overall absolute errors were greater than 439 persons/km<sup>2</sup> (the mean population density is 1470 persons/km<sup>2</sup>). The extreme low and high population-density block groups were the main sources of error. Low-population-density block groups had a more severe impact on relative errors, whereas high-population-density block groups had more impact on absolute errors. These



**FIGURE 10.2** Population-density distribution estimated using the best regression model (model 3) based on non-stratified categories.



**FIGURE 10.3** Residual distribution from model 4; negative indicates overestimated, and positive indicates underestimated.

impacts can be illustrated clearly in the scatter plot of the residuals. Figure 10.3 shows the residual distributions of the best model (model 3). It indicates that population in very low-density block groups was overestimated, whereas population in high-density area was greatly underestimated. The high estimation errors imply that no single model worked well for all levels of population density. In order to improve population-estimation results, separating the population density into subcategories such as low, medium, and high densities and developing models for each category become necessary.

#### 10.2.4 Result of Population Estimation Based on Stratified Sampling Method

Table 10.7 shows correlation coefficients between population parameters and remote sensing variables in the low-, medium-, and high-population-density categories. It is clear that in the low-density category, correlations were not as strong as those in medium- and high-density categories. Similar to the non-stratified scheme, in the medium- and high-density categories, temperature had the strongest positive correlation with population, whereas vegetation-related variables had negative correlations with population. The low correlation between remote sensing variables and population in the low-density category implies that population estimation for these areas was more complicated, and the issue warrants further study.

| Remote Sensing Variables |       | Low Density |         |         | Medium Density |         |         | High Density |         |         |
|--------------------------|-------|-------------|---------|---------|----------------|---------|---------|--------------|---------|---------|
|                          |       | PD          | SPD     | LPD     | PD             | SPD     | LPD     | PD           | SPD     | LPD     |
| ETM                      | B1    | -0.231†     | -0.237† | -0.232† | 0.398*         | 0.398*  | 0.39*   | 0.274†       | 0.269†  | 0.264†  |
|                          | B2    | -0.232†     | -0.234† | -0.230† | 0.340*         | 0.342*  | 0.338*  | 0.248†       | 0.243†  | 0.237†  |
|                          | B3    | -0.244†     | -0.245† | -0.237† | 0.349*         | 0.351*  | 0.346*  | 0.267†       | 0.263†  | 0.259†  |
|                          | B4    | 0.223       | 0.231†  | 0.207   | -0.354*        | -0.335* | -0.304* | -0.371*      | -0.371* | -0.371* |
|                          | B5    | -0.164      | -0.141  | -0.132  | -0.060         | -0.047  | -0.029  | -0.085       | -0.087  | -0.089  |
|                          | B7    | -0.256†     | -0.248† | -0.234† | 0.243*         | 0.247*  | 0.246*  | 0.194        | 0.191   | 0.188   |
| PCs                      | PC1   | -0.249†     | -0.247† | -0.237† | 0.302*         | 0.304*  | 0.302*  | 0.231        | 0.227   | 0.223   |
|                          | PC2   | 0.164       | 0.181   | 0.168   | -0.391*        | -0.373* | -0.347* | -0.379*      | -0.378* | -0.377* |
|                          | PC3   | -0.001      | 0.027   | 0.068   | -0.269*        | -0.291* | -0.316* | -0.019       | -0.009  | 0.001   |
| VIs                      | NDVI  | 0.253†      | 0.257†  | 0.237†  | -0.388*        | -0.376* | -0.354* | -0.346*      | -0.344* | -0.342* |
|                          | RDVI  | 0.255†      | 0.257†  | 0.242†  | -0.411*        | -0.409* | -0.398* | -0.318*      | -0.315* | -0.312* |
|                          | SAVI  | 0.253†      | 0.257†  | 0.237†  | -0.388*        | -0.377* | -0.354* | -0.347*      | -0.345* | -0.342* |
|                          | SVI   | 0.252†      | 0.256†  | 0.241†  | -0.392*        | -0.384* | -0.365* | -0.340*      | -0.337* | -0.335* |
|                          | RVI   | 0.211       | 0.210   | 0.191   | -0.514*        | -0.506* | -0.485* | -0.354*      | -0.353* | -0.351* |
|                          | TNDVI | 0.260†      | 0.266†  | 0.246†  | -0.320*        | -0.308* | -0.286* | -0.335*      | -0.333* | -0.330* |
| Frac.                    | GV    | 0.254†      | 0.258†  | 0.238†  | -0.388*        | -0.376* | -0.353* | -0.357*      | -0.356* | -0.355* |
|                          | IMP   | -0.273†     | -0.266† | -0.249† | 0.291*         | 0.290*  | 0.283*  | 0.264†       | 0.262†  | 0.260†  |

|       |          |                     |                     |        |                     |                     |                     |                    |                    |                    |
|-------|----------|---------------------|---------------------|--------|---------------------|---------------------|---------------------|--------------------|--------------------|--------------------|
| Text. | B3_3 × 3 | -0.119              | -0.124              | -0.129 | -0.047              | -0.025              | 0.001               | -0.123             | -0.125             | -0.127             |
|       | B7_3 × 3 | -0.149              | -0.143              | -0.133 | -0.150              | -0.138              | -0.123              | -0.130             | -0.132             | -0.133             |
|       | B3_5 × 5 | -0.104              | -0.111              | -0.125 | -0.133              | -0.114              | -0.090              | -0.103             | -0.105             | -0.107             |
|       | B7_5 × 5 | -0.142              | -0.137              | -0.132 | -0.218 <sup>†</sup> | -0.210 <sup>†</sup> | -0.198 <sup>†</sup> | -0.106             | -0.108             | -0.109             |
|       | B3_7 × 7 | -0.093              | -0.102              | -0.122 | -0.177              | -0.159              | -0.136              | -0.080             | -0.083             | -0.085             |
|       | B7_7 × 7 | -0.136              | -0.132              | -0.130 | -0.251 <sup>*</sup> | -0.244 <sup>*</sup> | -0.234 <sup>†</sup> | -0.076             | -0.078             | -0.080             |
| Temp  | TEMP     | -0.234 <sup>†</sup> | -0.231 <sup>†</sup> | -0.212 | 0.622 <sup>*</sup>  | 0.635 <sup>*</sup>  | 0.637 <sup>*</sup>  | 0.425 <sup>*</sup> | 0.423 <sup>*</sup> | 0.420 <sup>*</sup> |

\*Correlation at 99 percent confidence level (two-tailed).

<sup>†</sup>Correlation at 95 percent confidence level (two-tailed).

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**TABLE 10.7** Relationships between Population Parameters and Remote Sensing Variables Based on Stratified Samples

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The  $R^2$  values for individual regression models are summarized in Table 10.8. In the low- and high-density categories, the highest  $R^2$  were only 0.13 and 0.180, respectively. This indicates that the ETM+ data may not be suitable for population estimation in these categories. In the medium-density category, the combination of vegetation indices or vegetation-related variables and the incorporation of spectral response, textures, and temperature can provide good estimations, especially the latter, when  $R^2$  reached as high as 0.87, 0.86, and 0.83 for different forms of dependent variables. Overall, the transforms of population density did not significantly improve the estimation in these categories.

Table 10.9 displays the four best models, selected based on  $R^2$  and estimation errors. It shows that the results of population estimation for the low-density category using remote sensing variables were not satisfactory owing to high estimation errors. In the medium-density category, both models provided very good estimations using vegetation-related independent variables or using a combination of spectral, texture, and temperature variables. The  $R^2$  values reached 0.83 and 0.86, respectively, with a relative error of less than 29 percent and an absolute error of less than 384 in both models (compared with a mean value of 1417). Figure 10.4 illustrates estimated population-density distribution using the best regression model (model 7) for this category. For the high-density category, the model using temperature as the only independent variable yielded the best estimation result. The relative error was only 11.4 percent, but absolute error reached 429 (compared with a mean value of 3707). Overall, the performance of the estimation models was much improved after stratification of population density into three categories. This finding implies that stratification based on population density is necessary for developing population-estimation models using remotely sensed data.

Figure 10.5 shows the distribution of residuals when a combination of models 5, 7, and 8 was applied to predict the population of Indianapolis in 2000. Most underestimations and overestimations were located in the central part of the city. For example, block groups 1, 2, 3, 4, and 5 (marked in Fig. 10.5) with very high population densities were greatly underestimated. These block groups usually had several multistory apartment buildings for residential use. On the other hand, most overestimated block groups were found in the downtown area, where commercial or service uses dominated. For example, the most overestimated block group, block group 3, had a population density of 678 persons/km<sup>2</sup>, according to the Census data, but the estimated population density reached up to 1884 persons/km<sup>2</sup>. The second most overestimated area was observed in block group 7, where university and residential uses shared the land.

| Scale  | Variable   | PD                                         |       | SPD                                              |       | LPD                                              |       |
|--------|------------|--------------------------------------------|-------|--------------------------------------------------|-------|--------------------------------------------------|-------|
|        |            | Selected Var.                              | $R^2$ | Selected Var.                                    | $R^2$ | Selected Var.                                    | $R^2$ |
| Low    | Bands      | B7                                         | 0.066 | B7                                               | 0.062 | B3                                               | 0.056 |
|        | PCs        | PC1                                        | 0.062 | PC1                                              | 0.061 | PC1                                              | 0.056 |
|        | VIs or VRV | TNDVI                                      | 0.068 | TNDVI                                            | 0.071 | TNDVI                                            | 0.061 |
|        | Frac.      | IMP                                        | 0.075 | IMP                                              | 0.071 | IMP                                              | 0.062 |
|        | Temp.      | TEMP                                       | 0.055 | TEMP                                             | 0.053 | TEMP                                             | 0.045 |
|        | Mixture    | IMP, B3_7 $\times$ 7                       | 0.130 | IMP                                              | 0.071 | IMP                                              | 0.062 |
| Medium | Bands      | B1, B2                                     | 0.290 | B1, B2                                           | 0.283 | B1, B2, B4                                       | 0.290 |
|        | PCs        | PC2, PC3                                   | 0.277 | PC2, PC3                                         | 0.277 | PC2, PC3                                         | 0.269 |
|        | VIs        | TNDVI, SAVI, RDVI, SVI                     | 0.641 | TNDVI, SAVI                                      | 0.611 | TNDVI, SAVI                                      | 0.597 |
|        | Frac.      | GV                                         | 0.151 | GV                                               | 0.141 | GV                                               | 0.124 |
|        | Text.      | B7_7 $\times$ 7, B7_3 $\times$ 3           | 0.215 | B7_7 $\times$ 7, B7_3 $\times$ 3                 | 0.230 | B3_3 $\times$ 3, B3_5 $\times$ 5                 | 0.296 |
|        | Temp.      | TEMP                                       | 0.387 | TEMP                                             | 0.404 | TEMP                                             | 0.407 |
|        | VRV        | RVI, TNDVI, PC2, SAVI, RDVI, SVI           | 0.825 | RVI, TNDVI, PC2, SAVI, RDVI, SVI, B4             | 0.829 | TNDVI, PC2, RDVI, NDVI, SVI                      | 0.794 |
|        | B-temp.    | TEMP, B7                                   | 0.442 | TEMP, B7                                         | 0.461 | TEMP, B7                                         | 0.466 |
|        | Mixture    | B3_7 $\times$ 7, RVI, TNDVI, SAVI, B5, PC1 | 0.869 | TEMP, B3_7 $\times$ 7, RVI, TNDVI, SAVI, B5, PC1 | 0.863 | TEMP, B7_7 $\times$ 7, RVI, TNDVI, AVI, B5, RDVI | 0.831 |

**TABLE 10.8** Comparison of Regression Results for Different Population-Density Categories

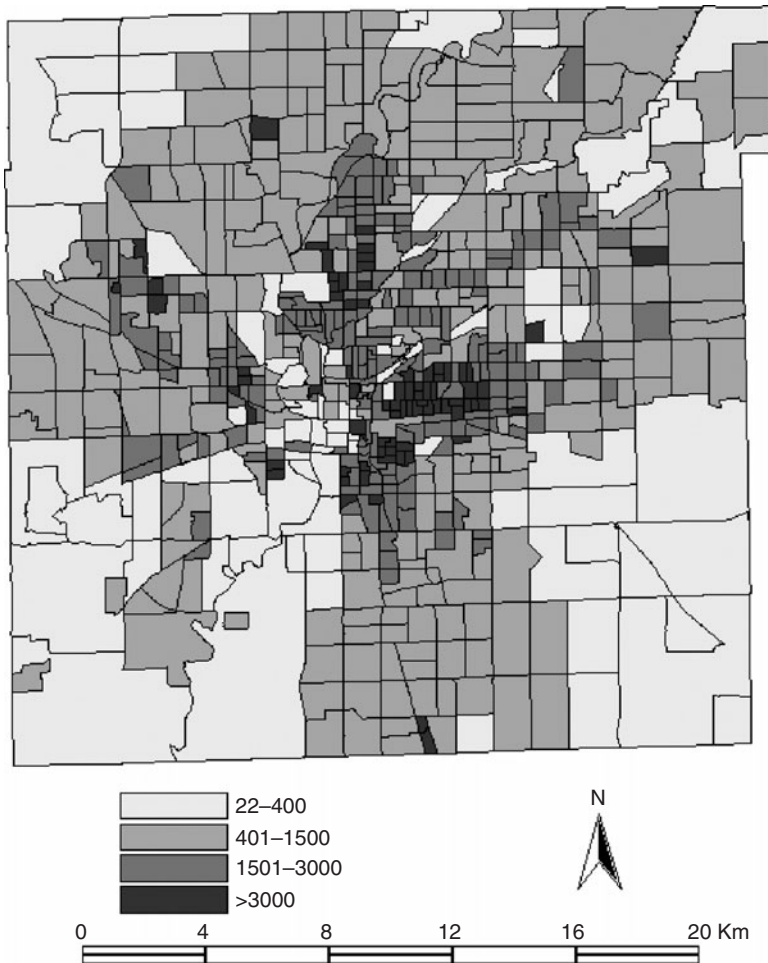
| Scale | Variable        | PD            |       | SPD           |       | LPD           |       |
|-------|-----------------|---------------|-------|---------------|-------|---------------|-------|
|       |                 | Selected Var. | $R^2$ | Selected Var. | $R^2$ | Selected Var. | $R^2$ |
| High  | Bands           | B4            | 0.138 | B4            | 0.138 | B4            | 0.138 |
|       | PCs/VRV         | PC2           | 0.144 | PC2           | 0.143 | PC2           | 0.142 |
|       | VIs             | RVI           | 0.125 | RVI           | 0.124 | RVI           | 0.123 |
|       | Frac.           | GV            | 0.128 | GV            | 0.127 | GV            | 0.126 |
|       | B-temp. mixture | TEMP          | 0.180 | TEMP          | 0.179 | TEMP          | 0.177 |

**TABLE 10.8** Comparison of Regression Results for Different Population-Density Categories (*Continued*)

| Model  |   | Potential Var.  | Dep. Var. | Regression Equation                                                                                                                                                                                                 | $R^2$ | RE    | AE  |
|--------|---|-----------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-----|
| Low    | 5 | Mixture         | PD        | $296.190 - 9.136 \times \text{IMP} + 7.746 \times \text{B3}_7 \times 7$                                                                                                                                             | 0.130 | 315.3 | 97  |
| Medium | 6 | VRV             | SPD       | $-649.707 + 37.317 \times \text{RVI} + 1027.726 \times \text{TNDVI} - 1.579 \times \text{PC2} - 538.330 \times \text{SAVI} + 0.608 \times \text{B4} - 12.092 \times \text{RDVI} + 1.183 \times \text{SVI}$          | 0.829 | 28.9  | 357 |
|        | 7 | Mixture         | SPD       | $-966.765 + 0.684 \times \text{TEMP} - 0.632 \times \text{B3}_7 \times 7 + 46.507 \times \text{RVI} + 1110.756 \times \text{TNDVI} - 581.426 \times \text{SAVI} - 1.003 \times \text{B5} + 0.514 \times \text{PC1}$ | 0.863 | 28.4  | 384 |
| High   | 8 | B-temp./Mixture | SPD       | $-565.332 + 2.047 \times \text{TEMP}$                                                                                                                                                                               | 0.179 | 11.4  | 429 |

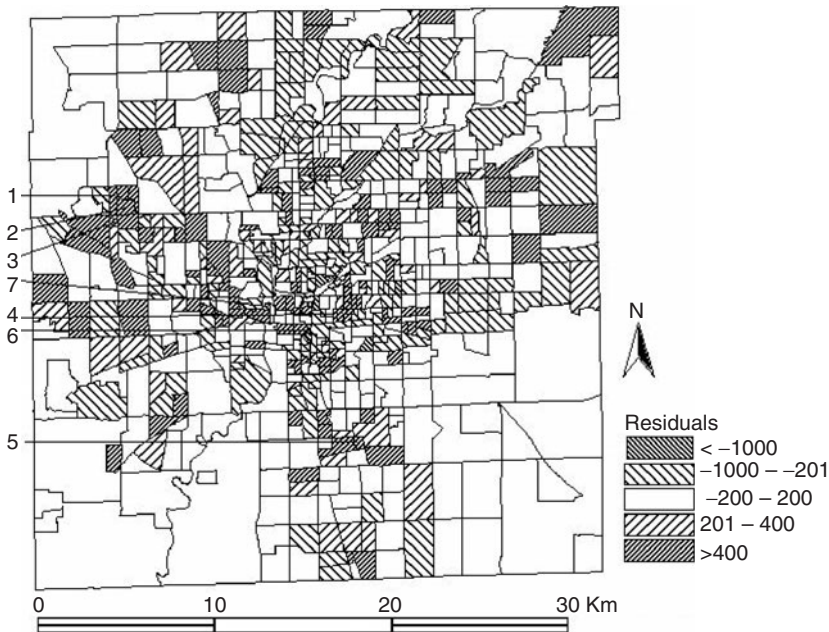
**TABLE 10.9** Best Estimation Models for Different Population-Density Categories Based on the Stratified Sampling Scheme





**FIGURE 10.4** Estimated population-density distribution using the best regression model (model 7) based on stratified category.

Based on the models developed for estimating population densities, the population of individual block groups can be calculated, and the total population of the whole city can be summed up. The total population estimates were 832,792 with a relative error of 3.2 percent using the stratified scheme (i.e., combination of models 5, 7, and 8) and 789,756 with a relative error of 8.2 percent using the non-stratified scheme (i.e., model 3). It is concluded that remote sensing techniques can provide reasonably accurate estimation results for the total population and that dividing population density into different categories is more effective than conventional non-stratified methods.



**FIGURE 10.5** Residual map of estimated population densities based on the best model of the stratified sampling scheme. Negative value indicates overestimation, and positive value indicates underestimation.

### 10.3 Discussion

Using remote sensing techniques to estimate population density is still a challenging task in terms of theory and methodology (owing to remotely sensed data per se), the complexity of urban landscapes, and the complexity of population distribution. Remotely sensed data are associated with the characteristics of surface features but not directly related to population counts or population densities. For example, the areas with low population density may be located in industrial/commercial areas or in forest/agricultural areas, but the spectral characteristics of these landscapes are fundamentally different. The areas with high population density generally have a number of multistory apartment buildings. The distribution of these high-density areas varies greatly with urban spatial structure, which has identified three classic types: concentric zone, sector, and the multiple-nuclei city (Wheeler and Muller, 1981). In developed countries, substantial urban growth frequently occurs in suburban areas owing to the redistribution of population and/or decentralization of metropolitan urban functions, whereas in developing countries, urban expansion is more related to rapid population growth and industrialization. In Eastern cities, such as Hong Kong, Singapore, and Shanghai,

high population density is often observed in the central part of the city, where high-rise residential buildings coalesce. Optical remote-sensing data, however, have not been linked directly to the vertical or internal features of such buildings for digital analysis of population estimation.

Remotely sensed data and census data are often collected with different formats and stored with different data structures. The values of any census variable are aggregated totals or mean values for the entire spatial extent of a census unit. In other words, census units are assumed to be homogeneous, no matter what types of variation in land use occur within them. A problem is created when unoccupied areas, such as water, airport, and forest, are given a population in the census. Remote sensing data have finer resolution than census data. When integrating these two types of data, a common method is to aggregate remotely sensed data to an appropriate census level. For example, this research aggregated various remote sensing variables at a block-group level. This aggregation has the potential to result in the same values for different block groups despite their differences in land-use and land-cover types, causing errors for population estimation.

Because of the complexity of population distribution, a single model is often difficult to fit all the data. Stratification of population density in this study has proved to be effective in improving estimation results. However, the method of stratifying population based on density may result in spatial discontinuity of the data, and it may be difficult to find suitable thresholds for stratification. The population categories identified here correspond, to some extent, to land-cover categories in urban residential areas in the National Land Cover Database (NLCD) (Vogelman et al., 1998, 2001). Low-intensity residential areas in the NLCD, mostly single-family housing units, relate to the medium-population-density category, whereas high-intensity residential areas, such as apartment complexes and row houses, relate to the high-population-density category. Moreover, given that various factors affect remotely sensed data quality, it is usually difficult to transfer the model developed in one site directly to other sites. Many factors need to be considered, including image acquisition date and time, the atmospheric condition when the image data were acquired, and the characteristics of the urban landscapes under investigation. Model transfer to nearby areas with similar socioeconomic conditions is favorable if these areas are within the same image scene or adjacent scenes in the same path of image acquisition.

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## 10.4 Conclusions

Population-estimation models developed based on the integration of satellite imagery and census data have numerous applications. They can be used to provide information on intra-urban population distribution,

which is essential in urban planning, natural-hazard risk assessment, disaster prevention and response, environmental impact assessment, transportation planning, economic decision making, and evaluation of quality of life. They also can be applied to validate urban growth models if the time-series image data become available.

This study demonstrates that Landsat ETM+ imagery could be used to provide reasonably accurate population-density estimation by combining various remote sensing-derived variables. Vegetation-related variables were especially effective. Remote sensing-based models were more suitable for estimation of population with medium-density than with low- and high-density areas. The stratification of population density into some categories and development of estimation models for individual categories improved model performance. Although population estimation using remotely sensed data is not straightforward, especially for low- and high-density regions, a major advantage of this approach is that it can provide a timely update of a population database and its spatial distribution, which is impossible by conventional census approaches.

More research is needed to improve population estimation through development of suitable models and use of multisource data, such as high spatial-resolution imagery and light detection and ranging (LiDAR) data, which are capable of identifying individual buildings and of measuring the heights of buildings. Comparative analyses of different methods are further suggested; for example, the method based on dwelling counts could be used to estimate population in low-density areas using high spatial-resolution remote sensing images. Further studies are warranted by incorporating Elvidge's radiance-calibrated nighttime lights imagery with original frequency data from satellite imagery because the former has been proven to be effective in estimating low- and high-population extremes (Elvidge et al., 1995, 1997). In addition, dynamic changes in population can be examined if multitemporal remote sensing and census data become available.

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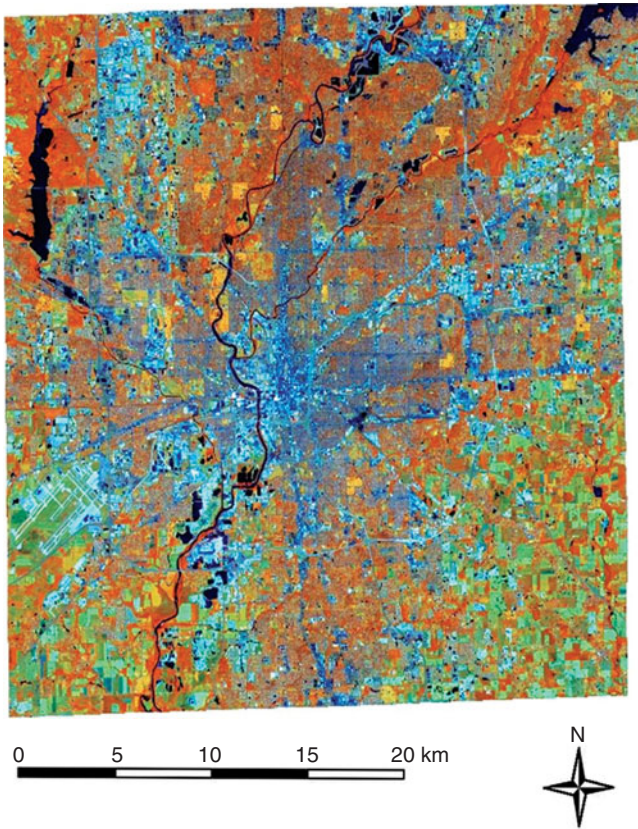
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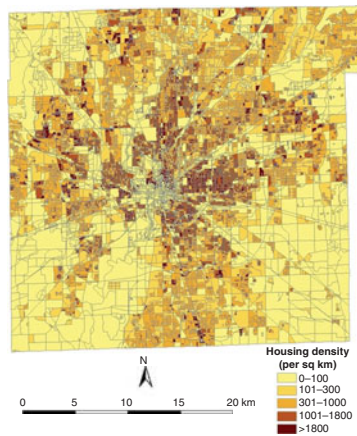
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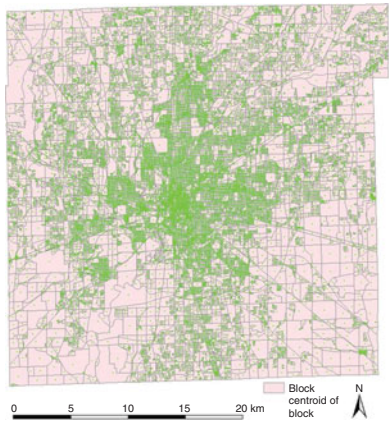


**FIGURE 3.2** Landsat ETM+ image (bands 4, 5, and 3 as red, green, and blue, respectively) of Indianapolis acquired on June 22, 2000.

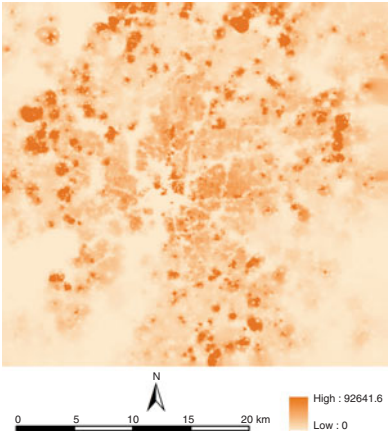


**FIGURE 3.4** Housing density distribution at Census block, Marion County, Indiana, in 2000.

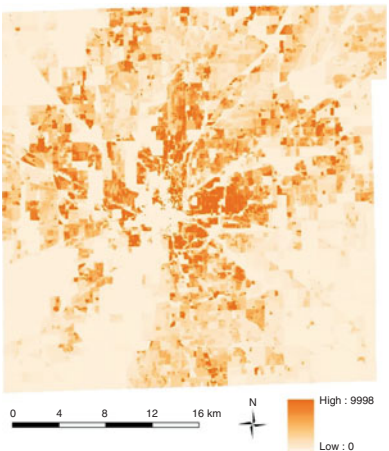
**FIGURE 3.5**  
Centroids of  
Census block.

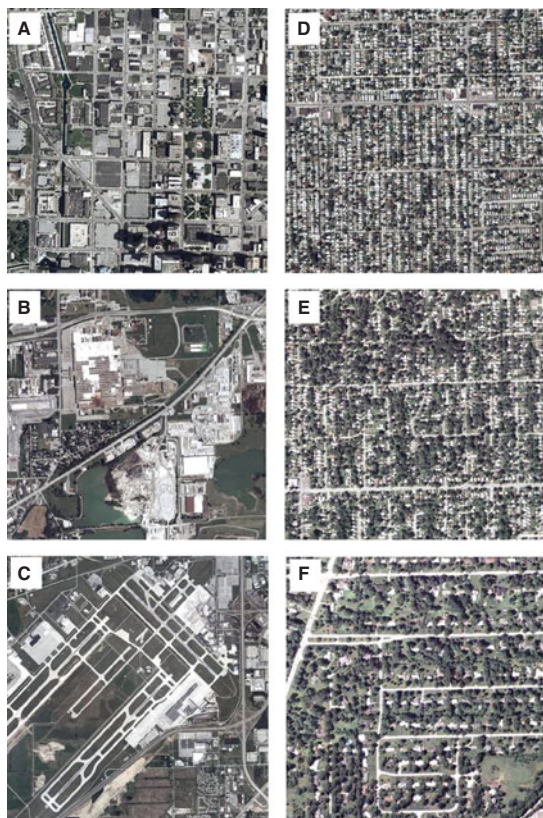


**FIGURE 3.6**  
Housing surface  
generated by IDW  
interpolation.

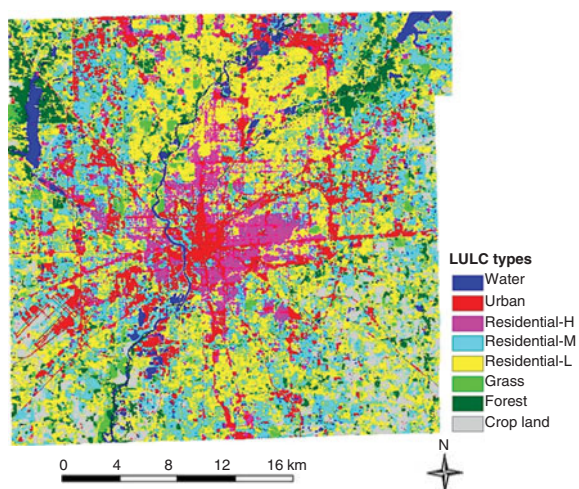


**FIGURE 3.7**  
Rasterized  
housing density.



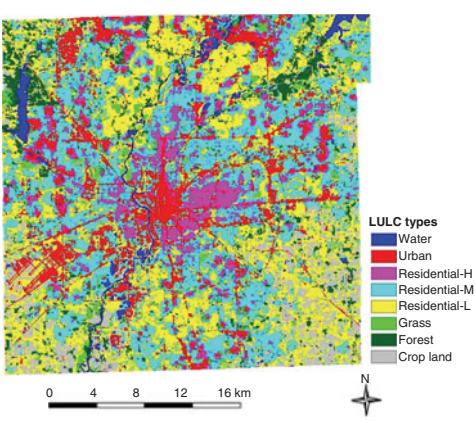


**FIGURE 3.8** Examples of typical land use: (a) commercial, (b) industrial, (c) transportation, (d) high-density residential, (e) medium-density residential, (f) low-density residential.

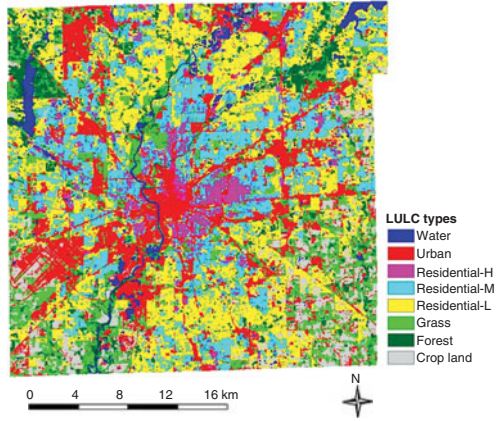


**FIGURE 3.9** LULC classification image based on intergration of housing and the ETM+ image at the pre-classification stage.

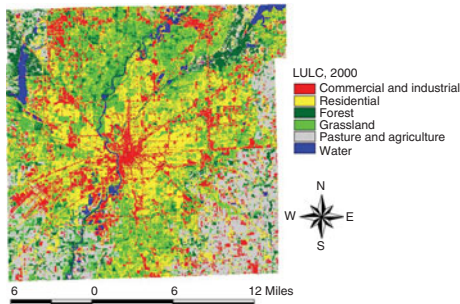
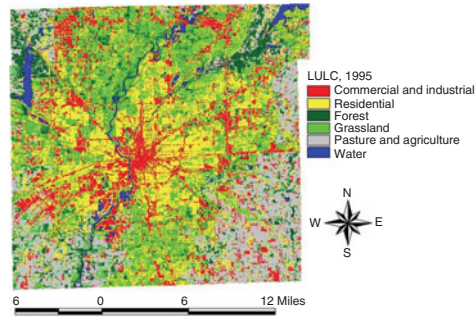
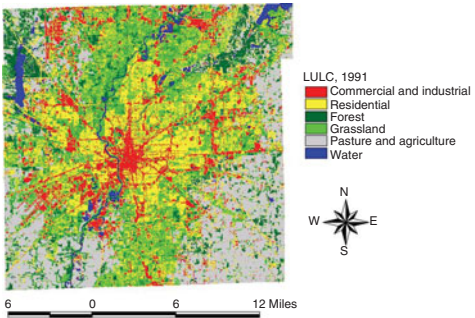




**FIGURE 3.10** LULC map based on integration of housing surface into the ETM+ image as an additional layer (during the classification).



**FIGURE 3.11** LULC map based on integration of housing data at the post-classification stage.



**FIGURE 4.6** LULC maps of 1991, 1995, and 2000. (Adapted from Weng and Lu, 2006.)